

Antibiotic Spectrum — Quick Reference

Ask 4 questions of any antibiotic: **GPC** (+MRSA/Enterococcus?) · **GNR** (+Pseudomonas/ESBL?) · **atypical** · **anaerobe** (+gut/B. fragilis?). Adapted from TeachIM (CC BY-NC 4.0); confirm vs local antibiogram.

Antibiogram grid

Agent	MRSA	Ent	PsA	ESBL	Atyp	Anae
Nafcillin/cefazolin	–	–	–	–	–	–
Amp-sulb / amox-clav	–	+	–	–	–	+
Pip-tazo	–	+	+	–	–	+
Ceftriaxone	–	–	–	–	–	–
Ceftazidime/cefepime	–	–	+	–	–	–
Ceftaroline (5th)	+	–	–	–	–	–
Ertapenem	–	–	–	+	–	+
Meropenem	–	~	+	+	–	+
Ciprofloxacin	–	–	+	–	+	–
Levofloxacin	–	~	+	–	+	~
Moxifloxacin	–	–	~	–	+	+
Aztreonam / aminoglyc.	–	–	+	~	–	–
Vanc/dapto/linezolid	+	+	–	–	–	–
Clinda/TMP-SMX/doxy	+	~	–	–	~	~

Escalation ladders

GPC / MRSA / Enterococcus: MSSA [nafcillin/cefazolin] → MRSA [vancomycin] → [daptomycin, *not lung*] → [linezolid, VRE & MRSA-PNA] → [ceftaroline].

GNR / Pseudomonas / ESBL: [ceftriaxone] → +PsA [ceftazidime/cefepime/pip-tazo] → ESBL [ertapenem (no PsA) / meropenem (+PsA)] → CRE/MDR (ID only) [ceftaz-avibactam, mero-vaborbactam, cefiderocol].

Severe PCN allergy → aztreonam (avoid if ceftazidime allergy) or aminoglycoside.

High-yield traps

- **Ertapenem** has NO Pseudomonas coverage.
- **No oral cephalosporin** covers Pseudomonas.
- **Daptomycin** inactivated by surfactant → never for pneumonia (toxicity: rhabdo).
- **Ceftazidime** poor CNS → not for meningitis.
- **Ciprofloxacin** poor Strep → not for respiratory infections.
- **Carbapenems** = best ESBL but poor Enterococcus.
- **Aztreonam** cross-reacts with ceftazidime (shared side chain).

Class rules & pearls

- **Penicillins:** no MRSA/atypical; GN rises Pen → antistaph → amino+BLI → pip-tazo (the one with PsA).
- **Cephalosporins:** no Enterococcus/atypical; GN rises by gen; ceftaroline (5th) = MRSA.
- **FQs:** cipro (urinary), levo (urinary+resp), moxi (resp, best anaerobe, weak PsA). Harms: QT, aneurysm, tendon, C. diff.
- **faecaLESS** = less resistant (amp-S, rarely VRE); **faecium** = more resistant (often VRE → linezolid/dapto).
- **Macrolides:** atypicals; rising Strep resistance. Fidaxomicin 1st-line for C. diff.

Newer agents (ID only)

- **Cefiderocol** — CRE, CRAB/Acinetobacter, Stenotrophomonas, PsA.
- **BL/BLI combos:** ceftazidime-avibactam, ceftolozane-tazobactam, meropenem-vaborbactam, imipenem-relebactam; aztreonam-avibactam (metallo-β-lactamases).
- **Long-acting anti-MRSA:** dalbavancin, oritavancin (single/2-dose ABSSI).